EEG Analysis Platform for Prestimulus EEG-Based Behavioral Prediction

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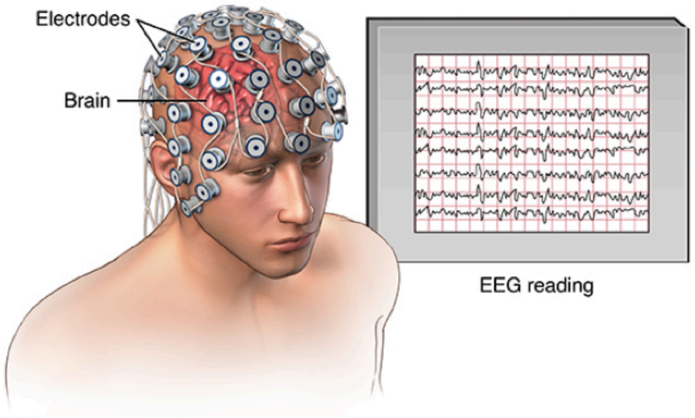
Brain–Computer Interface (BCI) and Electroencephalography (EEG)



Brain–Computer Interface (BCI)

A system that enables direct communication between the brain and an external device without relying on peripheral muscles.

Translates neural signals into control commands for interaction or decision-making.

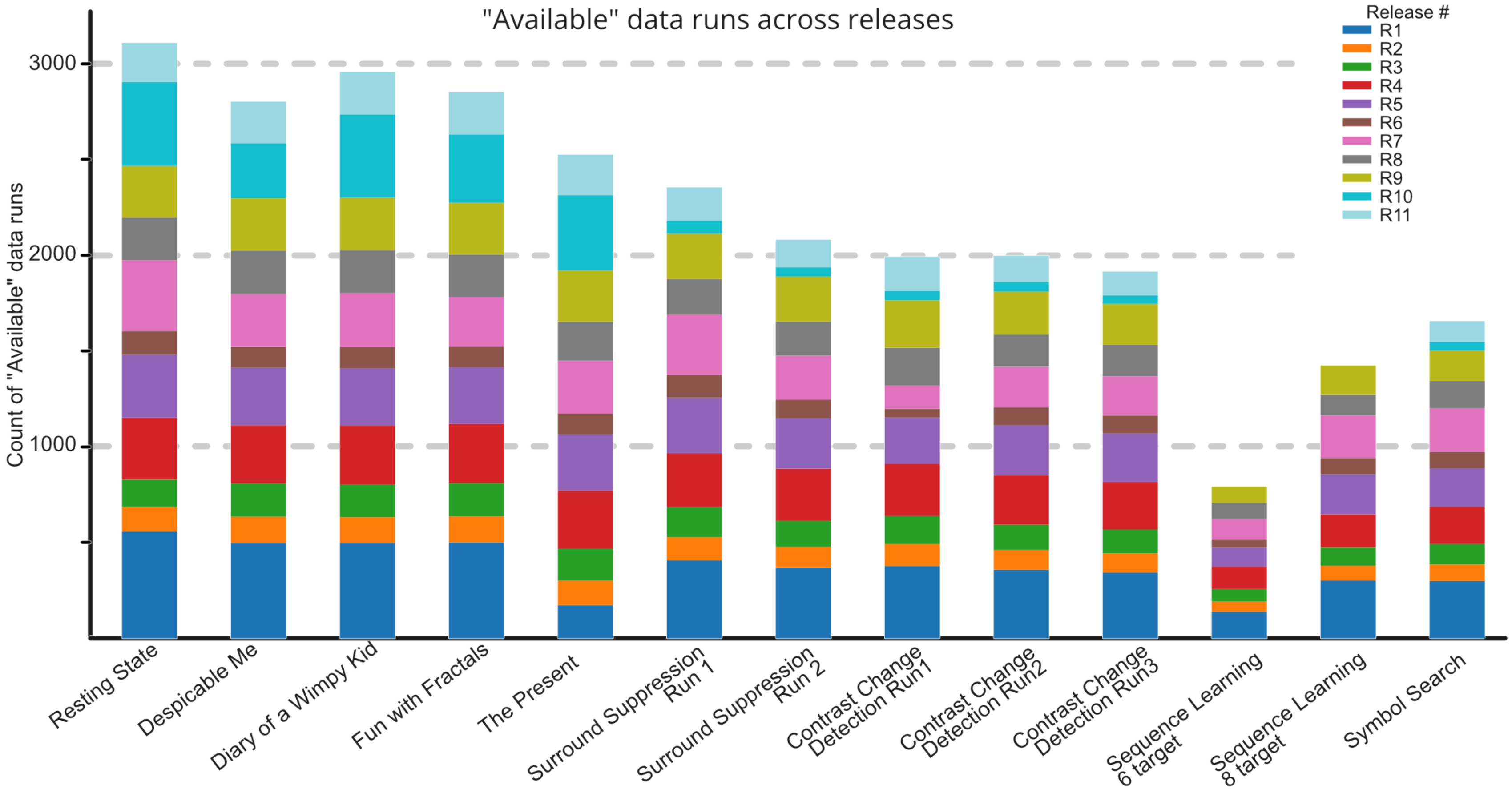
Electroencephalography (EEG)

A non-invasive technique for recording electrical activity generated by populations of cortical neurons.

Provides high temporal resolution, making it suitable for analyzing time-locked neural responses to events.

Why Reproducible EEG Analysis is Difficult

Many experimental tasks, high subject variability:



Key Challenge: As the number of experimental factors increases and subject variability grows, achieving reproducible and generalizable results becomes exponentially more difficult.

Practical Gap in EEG Research

Researchers usually choose between:

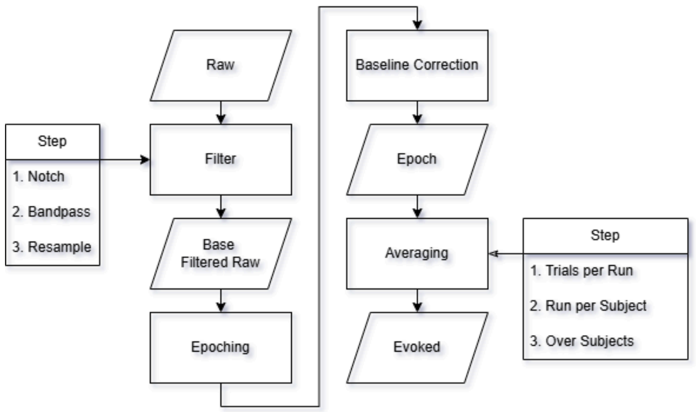
1. **Commercial tools** - easy to use, but limited and inflexible
2. **Custom scripts** - flexible, but require software engineering skills

Common outcome in practice:

- ad-hoc script copies (v1, v2, v3...)
- weak structure and little reuse
- hard-to-maintain and hard-to-reproduce pipelines

Impact: slower iteration, lower comparability across experiments, and harder collaboration.

Question: How can we provide a structured platform for researchers who can code but are not software engineers?



Research Objective

Investigate whether **prestimulus EEG segments** can predict:

1. **Reaction Time (RT)** - Regression target
2. **Trial Correctness** - Binary classification target

Dataset: Healthy Brain Network EEG (HBN-EEG)
Task: Contrast Change Detection (CCD)
Focus Window: 2-second inter-trial interval (ITI)

Solution Overview

EEG Analysis Platform - A systematic, reproducible analysis framework for domain experts

Design Philosophy:
Platform designed for researchers with coding capability but limited software engineering background

Key Principles:

- Systematic, reproducible workflows with explicit parameter management
- Modular architecture enabling independent component iteration
- Transparent experiment tracking and artifact caching
- Clear separation between domain logic and infrastructure

Target Use:

- Domain experts conducting EEG research
- Researchers needing efficient iteration on preprocessing and modeling
- Teams requiring standardized experimental protocols

Key Innovation: Integrate visualization-driven data validation with machine learning experimentation in a unified, parameter-driven interface

SECTION 1: PROJECT INTRODUCTION

Why This Problem Matters

Reproducibility Crisis in Neuroscience:

Despite EEG's widespread use, the research community lacks standardized analysis workflows. This creates:

- **Barrier to Reproducibility** - Experiments difficult to replicate or extend
- **Knowledge Fragmentation** - Each research group develops isolated solutions
- **High Cognitive Load** - Researchers must focus on infrastructure rather than science
- **Lost Efficiency** - Repeated development of similar functionality across labs

The Gap: Domain experts have domain knowledge but often lack professional software engineering training

Consequence: Code becomes difficult to maintain, test, and share—slowing scientific progress

Our Approach:
Design a platform that respects domain expertise while providing sound software engineering foundations

Our Solution: EEG Analysis Platform

Design Objectives:

1. **Systematic Workflows** - Deterministic, parameter-driven pipelines replacing ad-hoc scripts
2. **Reduced Cognitive Load** - Clear separation between domain logic and infrastructure
3. **Efficient Iteration** - Caching and modular design enabling rapid experimentation
4. **Transparent Reproducibility** - All parameters, seeds, and configurations explicitly logged
5. **Accessibility** - Intuitive interface for researchers with coding skills but limited SE background

Core Capabilities:

- Load & explore multi-subject BIDS-formatted EEG data
- Configure reproducible preprocessing pipelines with parameter versioning
- Generate comparison-ready datasets for train model
- Train and evaluate the EEGNet model systematically
- Export standardized results (metrics, plots, configurations)

Our Solution: EEG Analysis Platform

Success Criteria:

- Reduced time for experiment setup and iteration
- Increased reproducibility and experiment comparability
- Clear documentation of all processing decisions

Research Foundation: HBN-EEG Dataset

Healthy Brain Network (HBN) - Multi-Modal Neuroimaging Initiative

Dataset Characteristics:

- Large-scale, open-access dataset with diverse subjects
- Multi-task experimental paradigms with synchronized behavioral measurements
- BIDS-formatted structure enabling standardized analysis workflows
- High-density 128-channel EEG recordings

Complexity Considerations:

1. **Subject Heterogeneity** - Diverse demographics, developmental stages, and clinical backgrounds
2. **Multi-Task Design** - Multiple behavioral paradigms enabling diverse research questions
3. **High Dimensionality** - Temporal signals (>500 Hz sampling) across 128 channels = millions of data points per subject
4. **Data Quality Variability** - Artifacts, equipment issues, and subject factors affect signal quality
5. **Release Structure** - Multiple data releases with different processing levels and availability

Research Foundation: Why CCD Task

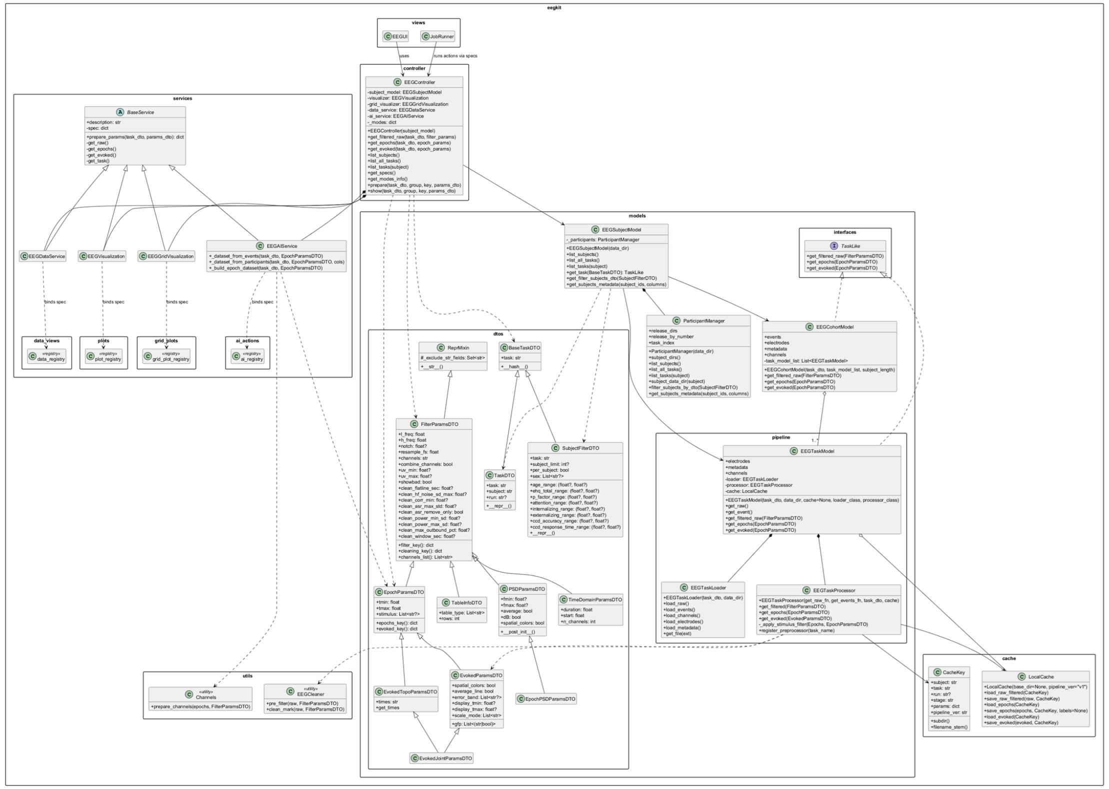
Strategic Focus: Contrast Change Detection (CCD) Task

Why CCD?

- **Dual Behavioral Targets:** Captures both accuracy (binary) and reaction time (continuous)
  - Enables validation of preprocessing and feature extraction
  - Provides comparison points across multiple modeling approaches
- **Clear Experimental Structure:** Well-controlled visual stimulus paradigm with unambiguous event markers
- **Sufficient Trial Density:** Adequate trials per subject for robust statistical modeling
- **Generalizability:** Serves as reference workflow applicable to other EEG tasks

Research Hypothesis:
Systematic analysis of prestimulus EEG activity during inter-trial intervals can reveal predictive neural signatures of subsequent behavioral performance

System Architecture Overview



- **Pipeline Pattern:** EEG processing workflow
- **Facade Pattern:** API endpoints
- **Strategy Pattern:** parameter-driven behavior
- **Factory Pattern:** model instantiation
- **Registry Pattern:** task preprocessor selection
- **Observer Pattern:** WebSocket progress updates
- **Repository Pattern:** structured EEG cache
- **Composite Pattern:** Pipeline Session is a combination of schemas

SECTION 3: SOFTWARE COMPLETION STATUS

Feature Completion Matrix

COMPLETED

Feature	Status	Evidence
Data loading & discovery	Complete	UC0 implemented, BIDS loader working
Signal inspection & visualization	Complete	Plots, topomaps, event tables generated
Preprocessing pipeline	Complete	Reproducible params stored, cached outputs
Epoch extraction	Complete	Deterministic trial alignment verified
Dataset building	Complete	Reusable training datasets created
Model training & evaluation	Complete	EEGNet trained and evaluated, metrics logged
Experiment logging & reproducibility	Complete	W&B tracking, config versioning
Backend API (FastAPI)	Complete	All endpoints functional
Frontend React UI	Complete	Modular architecture implemented and extensible for new workflows
Job/output storage system	Complete	Cache, jobs folders organized

INCOMPLETE / LIMITED

Feature	Status	Notes
WebSocket realtime updates	Partial	Basic structure; not fully integrated
Model performance	Needs work	See results section - underfitting & imbalance issues
Hyperparameter automation	Deferred	Manual tuning done; auto-tuning left
Cross-subject generalization	Limited	Subject-specific variance high; transfer learning not attempted
Real-time predictions	Deferred	Backend ready; inference optimization not priority

What Was Delivered

Software Artifacts

- **Backend codebase** (app/api, app/pipeline, app/ai_models, app/core, app/plots)
- **Frontend React UI** (modular mode-action design, extensible for future workflows)
- **Interactive analysis flows** (single-subject and cohort-filter execution)
- **Preprocessing pipeline** (BIDS loading, filtering, artifact removal, epoching)
- **Trained AI model (EEGNet)**
- **Experiment tracking** (W&B exports, metrics tables)
- **Reproducible notebooks** (eeg_workbench.ipynb)

What We Learned

- Preprocessed EEG data is tricky (many failure modes)
- Trial alignment bugs took longest to debug
- Class imbalance causes "accuracy illusion"
- Need validation before model iteration
- Caching intermediate results saves time massively

SECTION 4: RESULTS & EVALUATION

Manual Testing Results

Representative Functional Test Outcomes

Test Area	Example Check	Result
Input Configuration	Load valid single-subject EEG dataset	Pass
Cohort Filter	Apply valid subject/cohort filter	Pass
Preprocessing + Plot	Run filtering and generate frequency plot	Pass
Grid Plot Mode	Compare evoked responses across subjects	Pass
Data Mode	Inspect epochs table after segmentation	Pass
Caching	Re-run same config and reuse cache	Pass
AI Dataset Build	Construct training-ready tensors	Pass
Model Execution	Run EEG training and return metrics	Pass

Result Summary: Core user workflow was verified manually from data loading to model execution, with stable outputs under expected usage conditions.

Classification Results

Prediction Target: Is the trial correct or wrong?

Test Results Across Balancing Strategies:

Strategy	Accuracy	Balanced Acc	Sensitivity	Specificity
Baseline (no balance)	0.729	0.501	**0.993**	0.009
Class Weighting	0.604	0.544	0.674	0.414
Weighted Sampler	0.629	0.543	0.729	0.358
Undersampling	**0.652**	**0.537**	0.787	**0.287**

Interpretation:

- **Baseline shows "accuracy trap"**: achieves 72.9% by predicting *everything* as negative class
- **Class imbalance is severe** → minority class near-invisible to model
- **Best result**: Undersampling (0.652 acc) but still unstable
- **Problem**: No single strategy achieves robust, generalizable performance

Regression Results

Prediction Target: What will the reaction of hit target time be?

Test Results Across Loss Functions:

Loss Function	MAE	RMSE	R^2	Pearson
MAE	0.2876	0.3743	0.0063	0.0840
MSE	0.2883	0.3749	0.0030	0.0562
Huber	0.2874	**0.3740**	**0.0079**	**0.0903**

Interpretation:

- **R^2 ≈ 0** across all methods → model barely fits data
- **MAE ~0.29** → predictions close to overall mean RT
- **Huber performs marginally better** but still underfits
- **Problem**: Limited signal in prestimulus window OR inadequate feature engineering

What Went Wrong?

Known Challenges:

1. **High inter-subject variability** - EEG patterns differ across individuals
2. **Limited trial counts per subject** - Small sample size → overfitting risk
3. **Imbalanced correctness labels** - Most trials correct → model bias
4. **Artifact sensitivity** - Some subjects' recordings noisier than others
5. **Limited prestimulus info** - 2-second window before visual stimulus may inherently lack predictive signal

What We Tried:

- EEGNet model with multiple training/evaluation settings
- Various balancing strategies (class weights, resampling)
- Hyperparameter tuning
- Did NOT achieve robust predictive performance

Testing & Validation Protocol

What We Validated:

1. **Functional workflow tests** - Input config, preprocessing, plotting, caching, dataset build, and training flow
2. **Reproducibility checks** - Re-running identical configurations produced stable visual and tabular outputs
3. **No-leakage evaluation setup** - Subject-independent split strategy and consistent run configuration
4. **Failure-mode analysis** - Investigated imbalance collapse, weak specificity, and regression underfitting

Evidence Collected:

- Manual test report across Plot, Grid Plot, Data, and AI modes
- Metrics and artifacts in W&B exports
- Verification plots (SNR spectra, confusion matrices, prediction scatter plots)
- Logged preprocessing params, split definitions, and model configs

Lessons Learned

What Worked Well

- **Pipeline stability** - Preprocessing now deterministic & reproducible
- **Caching system** - Massive speedup when iterating on models
- **Modular architecture** - Easy to swap components
- **Comprehensive logging** - Can reproduce any run exactly

What Needs Improvement

- **Model generalization** - Current performance insufficient for practical use
- **Feature engineering** - Raw spectral features may need enhancement
- **Subject adaptation** - May need subject-specific models instead of universal
- **Data quality** - Some subjects' data cleaner than others

Future Work

1. Try transfer learning across subjects
2. Implement subject-specific models

Overall Project Assessment

Dimension	Status	Score
Reproducibility	Achieved	9/10
Software Quality	Good	8/10
Usability	Improved	7/10
Predictive Performance	Limited	3/10
Documentation	Complete	9/10

Verdict:

- **Strong engineering baseline** - Reproducible, well-architected platform
- **Limited AI performance** - Not yet production-ready for predictions
- **Research value** - Platform enables future iterations more efficiently

Key Takeaway

This project is a success as a research engineering effort:

- Replaced fragmented notebooks with reproducible pipeline
- Built systematic experimental workflow
- Identified specific failure modes & next steps
- Model performance remains an **open technical problem**

The platform is now ready for focused AI improvements.

Questions & Discussion

Ask us about:

- Live demonstration details
- Specific model architectures
- How you can extend this for your own research